

Exercises**Level - 1****(Problems Based on Fundamentals)****Area of a linear curve**

1. Find the area of the region bounded by the curves $y = x^2 - 2x + 2$, $x = -1$ and $x = 2$.
2. Find the area bounded by the curves $y = \ln x + \tan^{-1} x$ and the ordinates $x = 1$ and $x = 2$.
3. Find the area bounded by $y = x|\sin x|$, x-axis and the ordinates $x = 0$ and $x = 2\pi$.
4. Find the area bounded by $y = 1 \log_{\frac{1}{3}} x$ and the x-axis between $x = 1$ and $x = 2$.
5. Find the area of the region bounded by the curve $y = e^2x - 3e^x + 2$ and the x-axis.
6. Find the area bounded by $y = x^3$ and x-axis between the ordinates $x = -1$ and $x = 1$.
7. Find the area bounded by the curve $y = \sin x$ and the x-axis, for $0 \leq x \leq 2\pi$.
8. Find the area bounded by the curve $y = \cos x$ and the x-axis, for $0 \leq x \leq 2\pi$.
9. Find the area bounded by the curve $y = x(x - 1)(x - 2)$ and the x-axis.
10. Find the area bounded by the curve $y = (x - 1)(x - 2)(x - 3)$ and the x-axis.

Area of a region between two non-intersecting curves

11. Find the area of the region enclosed by the curves $y = x^2$ and $y = 2x - x^2$.

12. Find the area of the region bounded by the parabolas $y^2 = x$ and $x^2 = y$.
13. Find the area of the region bounded by the parabolas $4y^2 = 9x$ and $3x^2 = 16y$.
14. Find the area of the region bounded by the curves $y = \frac{3x^2}{4}$ and the line $3x - 2y + 12 = 0$.
15. Find the area bounded by the curve $x^2 = 4y$ and the straight line $x = 4y - 2$.
16. Find the area of the region included between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$.
17. Find the area of the region enclosed by the parabola $y^2 = 4ax$ and the chord $y = mx$.
18. Find the area of the region bounded by the curve $y = x^2 + 2$, $y = x$, $x = 0$ and $x = 3$.
19. Find the area bounded by the curve $y = 2x - x^2$ and the straight line $y = -x$.
20. Find the area bounded by the straight lines $x = 0$, $x = 2$ and the curves $y = 2x - x^2$, $y = 2^x$.
21. Find the area bounded by the curves $y = 6x - x^2$ and $y = x^2 - 2x$.
22. In what ratio does the x -axis divide the area of the region bounded by the parabolas $y = 4x - x^2$ and $y = x^2 - x$?
23. Find the area of the region between the circles $x^2 + y^2 = 4$ and $(x - 2)^2 + y^2 = 4$.
24. Find the area of the region enclosed between the two circles $x^2 + y^2 = 1$ and $(x - 1)^2 + y^2 = 1$.
25. Find the area of the smaller region bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the straight line $\frac{x}{a} + \frac{y}{b} = 1$.
26. Find the area of the region $\{(x, y): x^2 + y^2 \leq 1 \leq x + y\}$.
27. Find the area of the region $\{(x, y): y^2 \leq 4x, 4x^2 + 4y^2 \leq 9\}$.
28. Find the area of the region $\{(x, y): x^2 + y^2 \leq 2ax, y^2 \geq ax, x \geq 0, y \geq 0\}$.
29. Find the area of the region $\left\{ (x, y): \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 \leq \frac{x}{a} + \frac{y}{b} \right\}$.
30. Find the area of the region bounded by $y = 1 + |x + 1|$, $x = -2$, $x = 3$, $y = 0$.
31. Find the area of the region bounded by $y = 1 + |x + 1|$, $x = -3$, $x = 3$, $y = 0$.
32. Find the area enclosed by the curves $y = |x - 1|$ and $y = -|x - 1| + 1$.
33. Find the area of the region bounded by the curves $f(x) = \log x$ and $g(x) = (\log x)^2$.
34. Find the area enclosed by $y^2 = x^2 - x^4$.
35. Find the area bounded by the curve $|y| + \frac{1}{2} \leq e^{-|x|}$.
36. Find the area of the region $f(x, y) = [(x, y): 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, 0 \leq x \leq 2]$.
37. Find the area bounded by the curves $y = x$ and $y = x^3$.
38. Find the area bounded by the curves $y = x(x - 1)^2$ the y -axis and the line $y = 2$.
39. Find the area of the region bounded by $y = x^2 + 1$, $y = x$, $x = 0$ and $y = 2$.
40. Find the area of the region bounded by the curves $y = \log_e x$ and $y = 2^x$, $x = 1/2$ and $x = 2$.
41. Find the area enclosed by the parabola $(y - 2)^2 = (x - 1)$, the tangent to the parabola at $(2, 3)$ and x -axis.
42. Find the area enclosed by the loop of the curve $y^2 = (x - 1)(x - 2)^2$.
43. Find the area bounded by the curves $y = (x - 1)^2$, $y = (x + 1)^2$ and $y = 1/4$.
44. Find the area bounded by the curves $y = e^x$, $y = e^{-x}$ and the straight line $x = 1$.
45. Find the area bounded by the curve $y = \log x$, $y = \log |x|$, $y = |\log x|$ and $y = |\log |x||$.
46. Find the area bounded by the curve $|x - 2| + |y + 1| = 1$.
47. Find the area of the region bounded by the curve $[x] + [y] = 3$ in first quadrant, where $[.] = \text{GIF}$.
48. Find the area of the region bounded by the curve $|x| + |y| = 1$ and $|x - 1| + |y| = 1$.
49. Find the area bounded by the curve $y = \cos^{-1}(\cos x)$ and $|x - \pi| + \left| y - \frac{\pi}{2} \right| = \frac{\pi}{2}$.
50. Find the area bounded by the curve $|y| = -(1 - |x|)^2 + 5$.
51. Find the area bounded by the curves $y = \sqrt{|x|}$ and $y = |x|$.
52. Find the area of the region bounded by the curves $|x + y| + |x - y| \leq 2$.
53. Find the area bounded by the curve $y = x$ and $y = x + \sin x$, where $0 \leq x \leq \pi$.
54. Let $f(x)$ be a non-negative continuous function such that the area bounded by the curve $y = f(x)$, x -axis and the ordinates $x = \frac{\pi}{4}$ and $x = \beta > \frac{\pi}{4}$ is $\left(\beta \sin \beta + \frac{\pi}{4} \cos \beta + \sqrt{2} \beta \right)$, find $f\left(\frac{\pi}{2}\right)$.

Area of a region between two intersecting curves

55. Find the area bounded by the curves $y = \sin x$ and $y = \cos x$, $x = 0$ and $x = \frac{\pi}{2}$.
56. Find the area enclosed by the curves $y = 2 - |2 - x|$ and $y = \frac{3}{|x|}$.
57. The line $y = mx$ bisects the area enclosed by the lines $x = 0$, $y = 0$, $x = \frac{3}{2}$ and the curve $y = 1 + 4x - x^2$. Find m .
58. Find the area bounded by the curve $|y| + \frac{1}{2} \leq e^{-|x|}$.

Area of a region by a horizontal strip

59. Find the area bounded between $y = \sin^{-1} x$ and y -axis between $y = 0$ and $y = \pi/2$.
60. Find the area enclosed by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.
61. Find the area enclosed by $x = -2y^2$ and $x = 1 - 3y^2$.
62. Find the area enclosed by the curves $y = \tan^{-1} x$ and $y = \cot^{-1} x$ and the y -axis.
63. Find the area bounded by the curve $x = -y^2 + y + 2$ and the y -axis.

Area of a region between several graphs

64. Find the area of the plane figure bounded by $y = \sqrt{x}$, $x \in [0, 1]$, $y = x^2$, $x \in [1, 2]$ and $y = -x^2 + 2x + 4$, $x \in [0, 2]$.
65. Find the area of the region bounded by the curves $4|y| = |4 - x^2|$ and $|y|(x^2 + 4) = 12$.
66. Find the area bounded by the ellipse $x^2 + 2y^2 = 2$ and the outside of the parabola $y = 1 - x^2$.
67. Find the area common to the circle $x^2 + y^2 = 4$ and the ellipse $x^2 + 4y^2 = 9$.
68. Find the area enclosed by the curves $y = \ln(x + e)$ and $x = \ln\left(\frac{1}{y}\right)$, and x -axis.
69. Find the area enclosed by the curves $x^2 + y^2 = 4$, parabola $y = x^2 + x + 1$ and the curve $y = \left[\sin^2\left(\frac{x}{4}\right) + \cos\left(\frac{x}{4}\right)\right]$ and x -axis.
70. Find the area enclosed by the curves $\frac{x^2}{4} + y^2 = 1$ and $\frac{x^2}{2} - y^2 = 1$.

Least value of a variable area

71. If the area bounded by $y = \frac{x^3}{3} - x^2 + a$ and the straight lines $x = 0$ and $x = 2$ and the x -axis is minimum, find the value of a .
72. Find the value of a , for which the area bounded by the curve $y = a^2 x^2 + ax + 1$ and the straight lines $y = 0$, $x = 0$ and $x = 1$ the least.
73. For what value of k , the area enclosed by the curves $y = x^2 - 3$ and $y = kx + 2$ is the least. Find also the least area.
74. If the area bounded by the curve $y = x^2 + 2x - 3$ and the line $y = mx + 1$ is least, find the value of m .
75. Find the value of a for which the area bounded by the curve $y = \frac{x}{6} + \frac{1}{x^2}$, and the lines $y = 0$, $x = a$ and $x = 2a$ is least.

HINTS AND SOLUTIONS

Level I

1. Hence, the required area

$$\begin{aligned}
 &= \int_{-1}^2 y \, dx \\
 &= \int_{-1}^2 (x^2 - 2x + 2) \, dx \\
 &= \left(\frac{x^3}{3} - x^2 + 2x \right) \Big|_{-1}^2 \\
 &= \left(\frac{8}{3} - 4 + 4 + \frac{1}{3} + 1 - 2 \right) \\
 &= 3 + 1 - 2 = 2 \text{ sq.u.}
 \end{aligned}$$

2. Hence, the required area

$$\begin{aligned}
 &= \int_1^2 (\ln x + \tan^{-1} x) \, dx \\
 &= (x \log x - x) \Big|_1^2 + \left(x \tan^{-1} x - \frac{1}{2} \log |1 + x^2| \right) \Big|_1^2 \\
 &= (2 \log 2 - 1) + \left(2 \tan^{-1} 2 - \frac{1}{2} \log 5 - \frac{\pi}{4} \right) \\
 &= \left(2(\log 2 + \tan^{-1} 2) - \frac{1}{2} \log 5 - \left(\frac{\pi}{4} + 1 \right) \right) \text{ s.u.}
 \end{aligned}$$

3. Hence, the required area

$$\begin{aligned}
 &= \int_0^{2\pi} (x |\sin x|) \, dx \\
 &= \int_0^{\pi} (x \sin x) \, dx - \int_{\pi}^{2\pi} (x \sin x) \, dx \\
 &= (-x \cos x + \sin x) \Big|_0^{\pi} - (-x \cos x + \sin x) \Big|_{\pi}^{2\pi} \\
 &= \pi - (-2\pi - \pi) \\
 &= 4\pi \text{ sq.u.}
 \end{aligned}$$

4. Hence, the required area

$$\begin{aligned}
 &= \int_1^2 (\log_{1/2} x) \, dx \\
 &= -\frac{1}{\log 2} (x \log x - x) \Big|_1^2 \\
 &= -\frac{1}{\log 2} (2 \log 2 - 3) \\
 &= \frac{1}{\log 2} (3 - 2 \log 2) \text{ sq.u.}
 \end{aligned}$$

5. Hence, the required area

$$\begin{aligned}
 &= \int_0^{\log 2} (e^{2x} - 3e^x + 2) \, dx \\
 &= \left(\frac{e^{2x}}{2} - 3e^x + 2x \right) \Big|_0^{\log 2}
 \end{aligned}$$

$$\begin{aligned}
 &= \left(4 - 6 + 2 \log 2 + 3 - \frac{1}{2} \right) \\
 &= \left(\frac{1}{2} + 2 \log 2 \right) \text{ sq.u.}
 \end{aligned}$$

6. Hence, the required area

$$\begin{aligned}
 &= \int_{-1}^1 x^3 \, dx \\
 &= 0
 \end{aligned}$$

7. Hence, the required area

$$\begin{aligned}
 &= \int_0^{2\pi} \sin x \, dx \\
 &= \int_0^{\pi} \sin x \, dx - \int_{\pi}^{2\pi} \sin x \, dx \\
 &= (-\cos x) \Big|_0^{\pi} - (-\cos x) \Big|_{\pi}^{2\pi} \\
 &= -(\cos x) \Big|_0^{\pi} + (\cos x) \Big|_{\pi}^{2\pi} \\
 &= -(-1 - 1) + (1 - (-1)) = 4 \text{ sq.u.}
 \end{aligned}$$

8. Hence, the required area

$$\begin{aligned}
 &= \int_0^{2\pi} \cos x \, dx \\
 &= \int_0^{\pi/2} \cos x \, dx - \int_{\pi/2}^{3\pi/2} \cos x \, dx + \int_{3\pi/2}^{2\pi} \cos x \, dx \\
 &= (\sin x) \Big|_0^{\pi/2} - (\sin x) \Big|_{\pi/2}^{3\pi/2} + (\sin x) \Big|_{3\pi/2}^{2\pi} \\
 &= (1 - 0) - (-1 - 1) + (0 - (-1)) \\
 &= 4 \text{ sq.u.}
 \end{aligned}$$

9. Hence, the required area

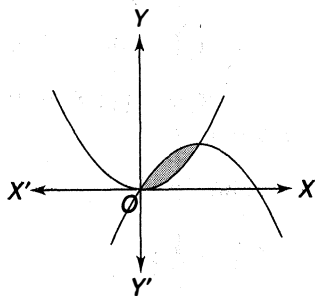
$$\begin{aligned}
 &= \int_0^2 \{x(x-1)(x-2)\} \, dx \\
 &= \int_0^1 \{x(x-1)(x-2)\} \, dx - \int_1^2 \{x(x-1)(x-2)\} \, dx \\
 &= \int_0^1 (x^3 - 3x^2 + 2x) \, dx - \int_1^2 (x^3 - 3x^2 + 2x) \, dx \\
 &= \left(\frac{x^4}{4} - x^3 + x^2 \right) \Big|_0^1 - \left(\frac{x^4}{4} - x^3 + x^2 \right) \Big|_1^2 \\
 &= \left(\frac{1}{4} - 1 + 1 \right) - \left((4 - 8 + 4) - \left(\frac{1}{4} - 1 + 1 \right) \right) \\
 &= 2 \left(\frac{1}{4} - 1 + 1 \right) \\
 &= \frac{1}{2} \text{ sq.u.}
 \end{aligned}$$

10. Hence, the required area

$$\begin{aligned}
 &= \int_0^3 \{(x-1)(x-2)(x-3)\} \, dx
 \end{aligned}$$

$$\begin{aligned}
 &= -\int_0^1 \{(x-1)(x-2)(x-3)\} dx \\
 &\quad + \int_1^2 \{(x-1)(x-2)(x-3)\} dx \\
 &\quad - \int_2^3 \{(x-1)(x-2)(x-3)\} dx \\
 &= -\left(\frac{x^4}{4} - 2x^3 + \frac{11}{2}x^2 - 6x\right)\Big|_0^1 \\
 &\quad + \left(\frac{x^4}{4} - 2x^3 + \frac{11}{2}x^2 - 6x\right)\Big|_1^2 \\
 &\quad - \left(\frac{x^4}{4} - 2x^3 + \frac{11}{2}x^2 - 6x\right)\Big|_2^3 \\
 &= -\left(\frac{1}{4} - 2 + \frac{11}{2} - 6\right) + (4 - 8 + 22 - 12) \\
 &\quad - \left(\frac{1}{4} - 2 + \frac{11}{2} - 6\right) \\
 &\quad - \left(\frac{81}{4} - 54 + \frac{99}{2} - 18\right) + (4 - 8 + 22 - 12) \\
 &= -2\left(\frac{23}{4} - 8\right) + 12 - \left(\frac{279}{4} - 72\right) \\
 &= \frac{46}{4} - \frac{279}{4} + 100 \\
 &= 100 - \frac{233}{4} = \frac{400 - 233}{4} = \frac{167}{4} \text{ sq.u.}
 \end{aligned}$$

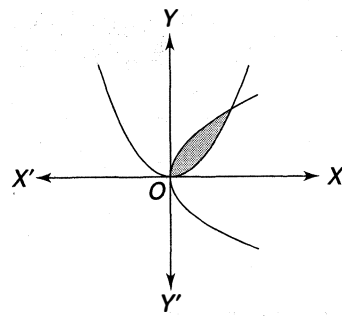
11. Given curves are $y = x^2$ and $y = 2x - x^2$



Hence, the required area

$$\begin{aligned}
 &= \int_0^1 (y_2 - y_1) dx \\
 &= \int_0^1 (2x - x^2 - x^2) dx \\
 &= \int_0^1 (2x - 2x^2) dx \\
 &= \left(x^2 - \frac{2}{3}x^3\right)\Big|_0^1 \\
 &= \left(1 - \frac{1}{3}\right) = \frac{2}{3} \text{ sq.u.}
 \end{aligned}$$

12. Given curves are $y^2 = x$ and $x^2 = y$.

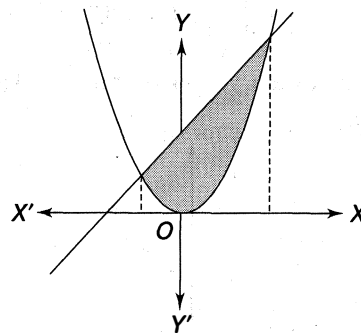


Hence, the required area

$$\begin{aligned}
 &= \int_0^1 (y_2 - y_1) dx \\
 &= \int_0^1 (\sqrt{x} - x^2) dx \\
 &= \left(\frac{2}{3}x^{3/2} - \frac{x^3}{3}\right)\Big|_0^1 \\
 &= \left(\frac{2}{3} - \frac{1}{3}\right) = \frac{1}{3} \text{ sq.u.}
 \end{aligned}$$

13. Do yourself.

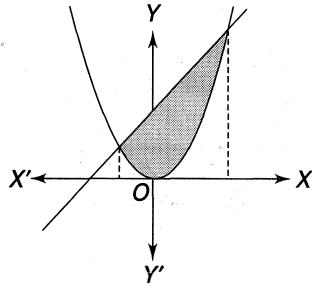
14. Given curves are $y = \frac{3x^2}{4}$ and $3x - 2y + 12 = 0$



Hence, the required area

$$\begin{aligned}
 &= \int_{-2}^4 (y_2 - y_1) dx \\
 &= \int_{-2}^4 \left(\frac{3x + 12}{2} - \frac{3x^2}{4}\right) dx \\
 &= \frac{3}{4} \int_{-2}^4 (2x + 8 - x^2) dx \\
 &= \frac{3}{4} \left(x^2 + 8x - \frac{x^3}{3}\right)\Big|_{-2}^4 \\
 &= \frac{3}{4} \left(\left(16 + 32 - \frac{54}{3}\right) - \left(4 - 16 + \frac{8}{3}\right)\right) \\
 &= \frac{3}{4} \left(36 - \frac{62}{3}\right) \\
 &= \frac{3}{4} \left(\frac{108 - 62}{3}\right) = \frac{46}{4} = \frac{23}{2} \text{ sq.u.}
 \end{aligned}$$

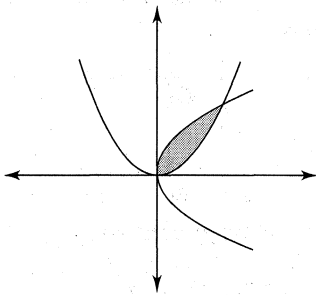
15. Given curves are $x^2 = 4y$ and $x = 4y - 2$



Hence, the required area

$$\begin{aligned}
 &= \int_{-1}^2 (y_2 - y_1) dx \\
 &= \int_{-1}^2 \left(\frac{x+2}{4} - \frac{x^2}{4} \right) dx \\
 &= \frac{1}{4} \int_{-1}^2 (x+2-x^2) dx \\
 &= \frac{1}{4} \left(\frac{x^2}{2} + 2x - \frac{x^3}{3} \right)_{-1}^2 \\
 &= \frac{1}{4} \left(\left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) \right) \\
 &= \frac{1}{4} \left(\frac{10}{3} + \frac{7}{6} \right) = \frac{1}{4} \left(\frac{27}{6} \right) = \frac{9}{8} \text{ sq.u.}
 \end{aligned}$$

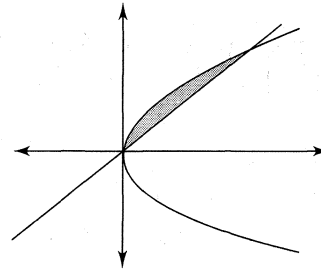
1 6. Given curves are $y^2 = 4ax$ and $x^2 = 4ay$.



Hence, the required area

$$\begin{aligned}
 &= \int_0^{4a} (y_2 - y_1) dx \\
 &= \int_0^{4a} \left(2\sqrt{a}\sqrt{x} - \frac{x^2}{4a} \right) dx \\
 &= 2\sqrt{a} \left(\frac{2}{3} x^{3/2} \right)_0^{4a} - \left(\frac{x^3}{12a} \right)_0^{4a} \\
 &= 2\sqrt{a} \left(\frac{2}{3} (4a)^{3/2} \right) - \left(\frac{64a^3}{12a} \right) \\
 &= \frac{32a^2}{3} - \frac{16a^2}{3} = \frac{16a^2}{3} \text{ sq.u.}
 \end{aligned}$$

1 7. Given curves are $y^2 = 4ax$ and $y = mx$.

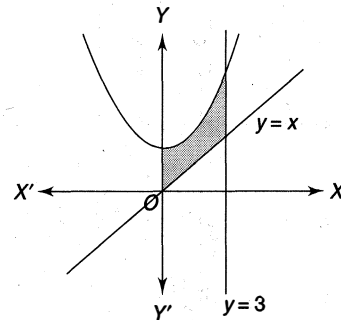


Hence, the required area

$$\begin{aligned}
 &= \int_0^{4a} (y_2 - y_1) dx \\
 &= \int_0^{4a} (2\sqrt{a}\sqrt{x} - mx) dx \\
 &= \left(2\sqrt{a} \left(\frac{2}{3} x^{3/2} \right) - m \left(\frac{x^2}{2} \right) \right)_0^{4a} \\
 &= \left(2\sqrt{a} \left(\frac{2}{3} (4a)^{3/2} \right) - m \left(\frac{16a^2}{2} \right) \right) \\
 &= \left(\frac{32}{3} - 8m \right) a^2 \text{ sq.u.}
 \end{aligned}$$

1 8. Given curves are

$$y = x^2 + 2, y = x, x = 0 \text{ and } x = 3.$$

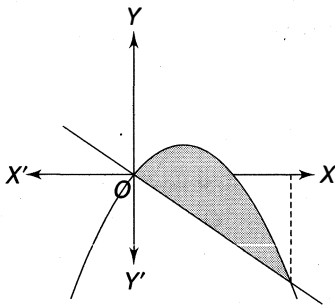


Hence, the required area

$$\begin{aligned}
 &= \int_0^3 (y_2 - y_1) dx \\
 &= \int_0^3 (x^2 + 2 - x) dx \\
 &= \left(\frac{x^3}{3} + 2x - \frac{x^2}{2} \right)_0^3 \\
 &= \left(\frac{27}{3} + 6 - 9 \right) = 6 \text{ sq.u.}
 \end{aligned}$$

1 9. Given curves are

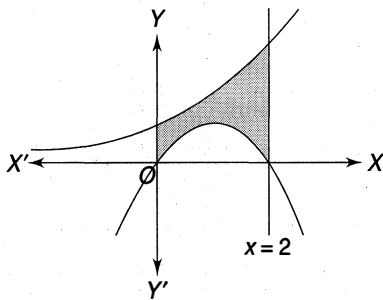
$$y = 2x - x^2 \text{ and } y = -x.$$



Hence, the required area

$$\begin{aligned} &= \int_0^3 (y_2 - y_1) dx \\ &= \int_0^3 (2x - x^2 + x) dx \\ &= \int_0^3 (3x - x^2) dx \\ &= \left(\frac{3x^2}{2} - \frac{x^3}{3} \right)_0^3 \\ &= \left(\frac{27}{2} - 9 \right) = \frac{9}{2} \text{ sq.u.} \end{aligned}$$

20. Given curves are $y = 2x - x^2$, $y = 2^x$ and $x = 0$, $x = 2$

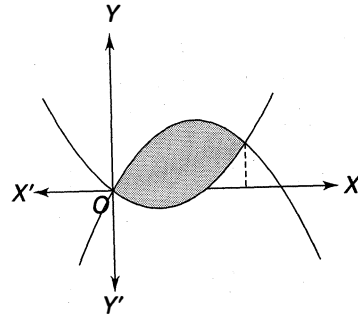


Hence, the required area

$$\begin{aligned} &= \int_0^2 (y_2 - y_1) dx \\ &= \int_0^2 (2x - x^2 - 2^x) dx \\ &= \left(x^2 - \frac{x^3}{3} - \frac{2^x}{\log 2} \right)_0^2 \\ &= \left(4 - \frac{8}{3} - \frac{4}{\log 2} + \frac{1}{\log 2} \right) \\ &= \left(\frac{4}{3} - \frac{3}{\log 2} \right) \text{ sq.u.} \end{aligned}$$

21. Given curves are

$$y = 6x - x^2 \text{ and } y = x^2 - 2x.$$



We have $6x - x^2 = x^2 - 2x$.

$$2x^2 = 8x$$

$$x = 0, 4$$

Hence, the required area

$$\begin{aligned} &= \int_0^4 (y_2 - y_1) dx \\ &= \int_0^4 (6x - x^2 - x^2 + 2x) dx \\ &= \int_0^4 (8x - 2x^2) dx \\ &= \left(4x^2 - \frac{2}{3} x^3 \right)_0^4 \\ &= \left(64 - \frac{128}{3} \right) = \frac{64}{3} \text{ sq.u.} \end{aligned}$$

22. Let A_1 be the area of the curve $y = 4x - x^2$ with x -axis and A_2 be the area of the curve $y = x^2 - x$ with x -axis

$$\begin{aligned} A_1 &= \int_0^4 y dx \\ &= \int_0^4 (4x - x^2) dx \\ &= \left(2x^2 - \frac{x^3}{3} \right)_0^4 \\ &= \left(32 - \frac{64}{3} \right) = \frac{32}{3} \end{aligned}$$

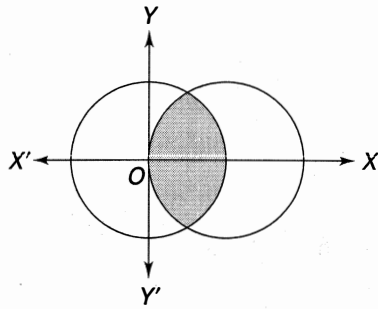
$$\begin{aligned} \text{and } A_2 &= \int_0^1 y dx \\ &= \int_0^1 (x - x^2) dx \\ &= \left(\frac{x^2}{2} - \frac{x^3}{3} \right)_0^1 \\ &= \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{6} \end{aligned}$$

Hence, the required ratio,

$$\frac{A_1}{A_2} = \frac{32}{3} \div \frac{1}{6} = \frac{32}{3} \times \frac{6}{1} = 64 : 1$$

23. Given curves are

$$x^2 + y^2 = 4 \text{ and } (x - 2)^2 + y^2 = 4.$$

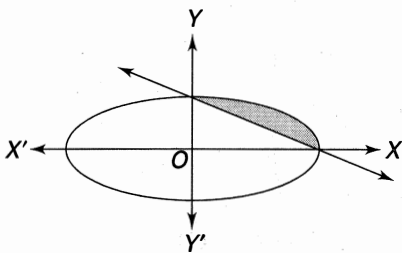


Hence, the required area

$$\begin{aligned} &= 2 \left(\int_0^1 \sqrt{4 - (x-2)^2} dx + \int_1^2 \sqrt{4 - x^2} dx \right) \\ &= 2 \left(\left(\frac{(x-2)}{2} \sqrt{4 - (x-2)^2} + \frac{4}{2} \sin^{-1} \left(\frac{x-2}{2} \right) \right) \Big|_0^1 \right. \\ &\quad \left. + 2 \left(\frac{x}{2} \sqrt{4 - x^2} + \frac{4}{2} \sin^{-1} \left(\frac{x}{2} \right) \right) \Big|_1^2 \right) \\ &= 2 \left(-\frac{\sqrt{3}}{2} + 2 \sin^{-1} \left(-\frac{1}{2} \right) - 2 \sin^{-1}(-1) \right) \\ &\quad + 2 \left(\sin^{-1}(1) - \frac{\sqrt{3}}{2} - 2 \sin^{-1} \left(\frac{1}{2} \right) \right) \\ &= 2 \left(-\frac{\sqrt{3}}{2} - \frac{\pi}{3} + \pi + \frac{\pi}{2} - \frac{\sqrt{3}}{2} - \frac{\pi}{3} \right) \\ &= 2 \left(\frac{3\pi}{2} - \frac{2\pi}{3} - \sqrt{3} \right) \\ &= \left(\frac{5\pi}{3} - 2\sqrt{3} \right) \text{ sq.u.} \end{aligned}$$

24. Do yourself.

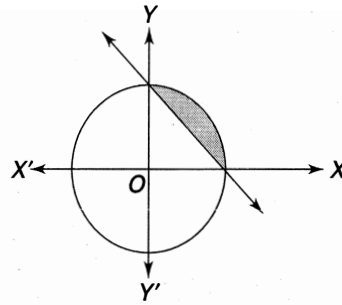
25. Given curves are $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $\frac{x}{a} + \frac{y}{b} = 1$.



Hence, the required area

$$\begin{aligned} &= \frac{\pi ab}{4} - \frac{1}{2} ab \\ &= \frac{1}{2} \left(\frac{\pi}{2} - 1 \right) \text{ sq.u.} \end{aligned}$$

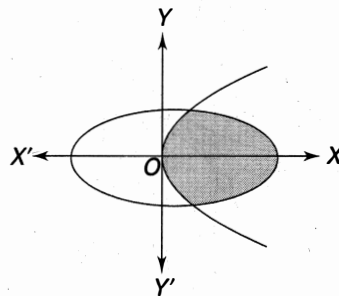
26. Given curves are $\{(x, y) : x^2 + y^2 \leq 1 \leq x + y\}$.



Hence, the required area

$$\begin{aligned} &= \int_0^1 (C - L) dx \\ &= \int_0^1 (\sqrt{1 - x^2} - (1 - x)) dx \\ &= \left(\frac{x\sqrt{1 - x^2}}{2} + \frac{1}{2} \sin^{-1} \left(\frac{x}{2} \right) - x + \frac{x^2}{2} \right) \Big|_0^1 \\ &= \left(\frac{1}{2} \sin^{-1} \left(\frac{1}{2} \right) - 1 + \frac{1}{2} \right) \\ &= \left(\frac{1}{2} \sin^{-1} \left(\frac{1}{2} \right) - \frac{1}{2} \right) \\ &= \frac{1}{2} \left(\frac{\pi}{6} - 1 \right) \text{ sq.u.} \end{aligned}$$

27. Given curves are $\{x : y^2 \leq 4x, 4x^2 + 4y^2 \leq 9\}$.



Solving $y^2 = 4x$ and $4x^2 + 4y^2 = 9$ we get.

$$\begin{aligned} &4x^2 + 16x - 9 = 0 \\ \Rightarrow &4x^2 - 2x + 18x - 9 = 0 \\ \Rightarrow &2x(2x - 1) + 9(2x - 1) = 0 \\ \Rightarrow &(2x + 9)(2x - 1) = 0 \\ \Rightarrow &x = \frac{1}{2}, -\frac{9}{2} \end{aligned}$$

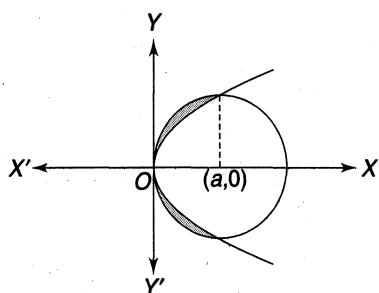
Hence, the required area

$$\begin{aligned} &= \int_0^{1/2} P dx + \int_{1/2}^{3/2} E dx \\ &= \int_0^{1/2} 2\sqrt{x} dx + \frac{1}{2} \int_{1/2}^{3/2} \sqrt{9 - 4x^2} dx \\ &= \int_0^{1/2} 2\sqrt{x} dx + \frac{1}{2} \int_{1/2}^{3/2} \sqrt{\left(\frac{3}{2}\right)^2 - x^2} dx \end{aligned}$$

$$\begin{aligned}
 &= \left(2 \times \frac{2}{3} \times x^{2/3}\right)_{0}^{1/2} + \frac{1}{2} \left(\frac{x}{2} \sqrt{\left(\frac{3}{2}\right)^2 - x^2}\right) \\
 &\quad + \frac{9}{8} \sin^{-1}\left(\frac{2x}{3}\right)_{1/2}^{3/2} \\
 &= \frac{4}{3} \times \frac{1}{2\sqrt{2}} + \frac{9}{8} \times \frac{1}{2} \sin^{-1}(1) \\
 &\quad - \frac{1}{2} \left(\frac{\sqrt{2}}{4} + \frac{9}{8} \sin^{-1}\left(\frac{1}{3}\right)\right) \\
 &= \frac{2}{3\sqrt{2}} + \frac{9\pi}{32} - \frac{1}{4\sqrt{2}} - \frac{9}{16} \sin^{-1}\left(\frac{1}{3}\right) \\
 &= \left(\frac{5}{12\sqrt{2}} + \frac{9\pi}{32} - \frac{9}{16} \sin^{-1}\left(\frac{1}{3}\right)\right) \text{ sq.u.}
 \end{aligned}$$

28. Given curves are

$$\{x: x^2 + y^2 \leq 2ax, y^2 \geq ax, x \geq 0, y \geq 0\}$$



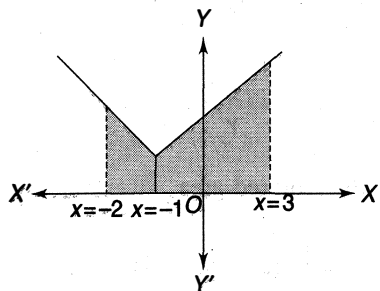
Hence, the required area

$$\begin{aligned}
 &= \int_0^a (y_2 - y_1) dx \\
 &= \int_0^a (\sqrt{a^2 - (x-a)^2} - a\sqrt{x}) dx \\
 &= \left(\frac{x-a}{2} \sqrt{a^2 - (x-a)^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x-a}{a}\right) - \frac{2a}{3} x^{3/2}\right)_0^a \\
 &= \frac{2a}{3} a\sqrt{a} - \frac{a^2}{2} \sin^{-1}(-1) \\
 &= \left(\frac{2a^{5/2}}{3} + \frac{\pi a^2}{4}\right) \text{ sq.u.}
 \end{aligned}$$

29. Do yourself.

30. Given curves are

$$y = 1 + |x + 1|, x = -2, x = 3, y = 0.$$



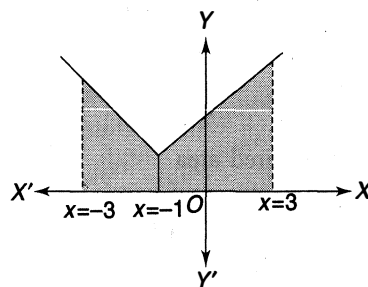
Hence, the required area

$$= \frac{1}{2} \times (2+1) \times 1 + \frac{1}{2} \times (1+2) \times 1 + \frac{1}{2} \times (2+5) \times 3$$

$$= \frac{1}{2} (3 + 2 + 21) = \frac{27}{2} \text{ sq.u.}$$

31. Given curves are

$$y = 1 + |x + 1|, x = -3, x = 3, y = 0.$$

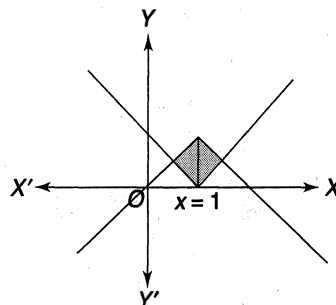


Hence, the required area

$$\begin{aligned}
 &= \frac{1}{2} \times (3+1) \times 2 + \frac{1}{2} \times (1+2) \times 1 + \frac{1}{2} \times (2+5) \times 3 \\
 &= \frac{1}{2} (4 + 3 + 21) = \frac{28}{2} = 14
 \end{aligned}$$

32. Given curves are

$$y = |x - 1| \text{ and } y = -|x - 1| + 1.$$

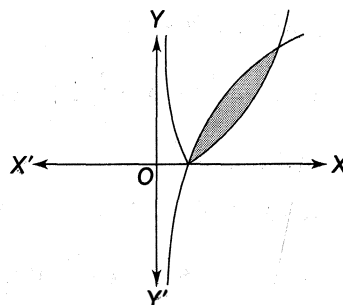


Hence, the required area

$$= 2 \left(\frac{1}{2} \times 1 \times \frac{1}{2}\right) = \frac{1}{2} \text{ sq.u.}$$

33. Given curves are

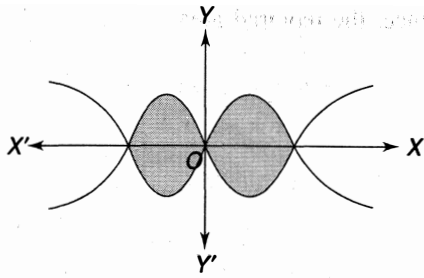
$$f(x) = \log x \text{ and } g(x) = (\log x)^2$$



Hence, the required area

$$\begin{aligned}
 &= \int_1^e (\log x - (\log x)^2) dx \\
 &= (x \log x - x - (x(\log x)^2 - 2(x \log x - x)))_1^e \\
 &= ((3(x \log x - x) - x(\log x)^2))_1^e \\
 &= (3 - e) \text{ sq.u.}
 \end{aligned}$$

34. Given curve is $y^2 = x^2 - x^4$.

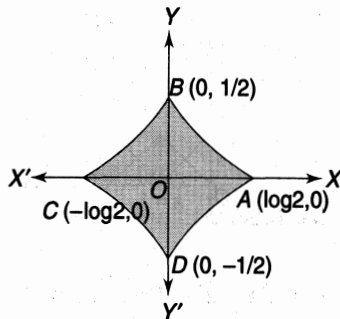


Hence, the required area

$$\begin{aligned} &= 4 \int_0^1 (x^2 - x^4) dx \\ &= 4 \left(\frac{x^3}{3} - \frac{x^5}{5} \right)_0^1 \\ &= 4 \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{8}{15} \text{ sq.u.} \end{aligned}$$

35. Given curve is

$$|y| + \frac{1}{2} \leq e^{-|x|}$$

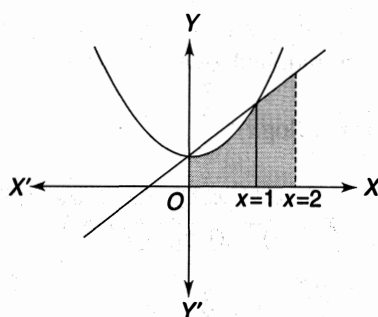


Hence, the required area

$$\begin{aligned} &= 4 \left(\int_0^{\log 2} e^{-x} dx \right) \\ &= 4 \left(-e^{-x} \right)_0^{\log 2} \\ &= 4 \left(1 - e^{-\log 2} \right) \\ &= 4 \left(1 - \frac{1}{2} \right) = 4 \times \frac{1}{2} = 2 \text{ sq.u.} \end{aligned}$$

36. Given curve is

$$f(x, y) = [(x, y) : 0 \leq y \leq x^2 + 1, \\ 0 \leq y \leq x + 1, 0 \leq x \leq 2]$$

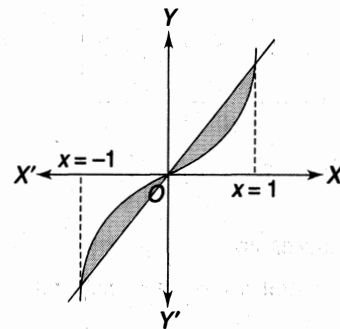


Hence, the required area

$$\begin{aligned} &= \int_0^1 (x^2 + 1) dx + \int_1^2 (x + 1) dx \\ &= \left(\frac{x^3}{3} + x \right)_0^1 + \left(\frac{x^2}{2} + x \right)_1^2 \\ &= \left(\frac{1}{3} + 1 \right) + \left(4 - \frac{3}{2} \right) \\ &= \frac{4}{3} + \frac{5}{3} = 3 \text{ sq.u.} \end{aligned}$$

37. Given curves are

$$y = x \text{ and } y = x^3.$$

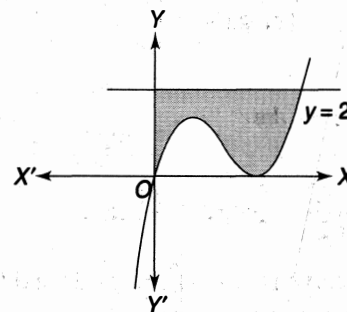


Hence, the required area

$$\begin{aligned} &= 2 \int_0^1 (x - x^3) dx \\ &= 2 \left(\frac{x^2}{2} - \frac{x^4}{4} \right)_0^1 \\ &= 2 \left(\frac{1}{2} - \frac{1}{4} \right) = 2 \times \frac{1}{4} = 1 \text{ sq.u.} \end{aligned}$$

38. Given curve is

$$y = x(x - 1)$$



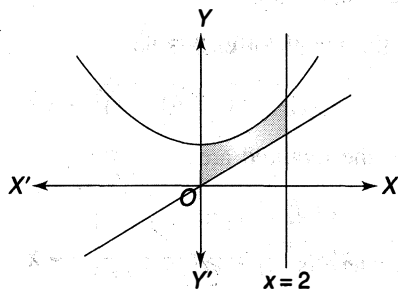
Hence, the required area

$$\begin{aligned} &= \int_0^2 (2 - x(x - 1)^2) dx \\ &= \int_0^2 (2 - x(x^2 - 2x + 1)) dx \end{aligned}$$

$$\begin{aligned}
 &= \int_0^2 (2 - (x^3 - 2x^2 + x)) dx \\
 &= \left(2x - \frac{x^4}{4} + \frac{2}{3}x^3 - \frac{x^2}{2} \right)_0^2 \\
 &= \left(4 - 4 + \frac{16}{3} - 2 \right) \\
 &= \frac{10}{3} \text{ sq.u.}
 \end{aligned}$$

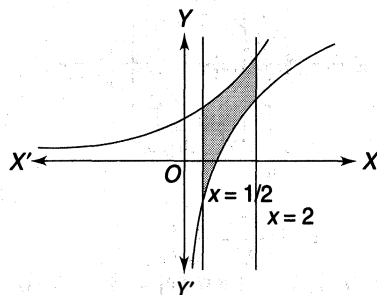
39. Given curves are

$$y = x^2 + 1, y = x, x = 0 \text{ and } y = 2.$$



40. Given curves are

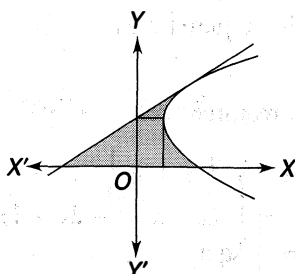
$$y = \log_e x \text{ and } y = 2^x, x = 1/2 \text{ and } x = 2.$$



Hence, the required area

$$\begin{aligned}
 &= \int_{1/2}^2 (2^x - \log x) dx \\
 &= \left(\frac{2^x}{\log 2} - (x \log x - x) \right)_{1/2}^2 \\
 &= \left(\frac{4}{\log 2} - 21 \log 2 + 2 - \frac{\sqrt{2}}{\log 2} + \frac{1}{2} \log \left(\frac{1}{2} \right) - \frac{1}{2} \right) \\
 &= \left(\frac{4 - \sqrt{2}}{\log 2} - \frac{5}{2} \log 2 + \frac{3}{2} \right) \text{ sq.u.}
 \end{aligned}$$

41. Given curve is $(y - 2)^2 = (x - 1)$ and the equation of the tangent to the curve at $(2, 3)$ is $x - 2y + 4 = 0$.

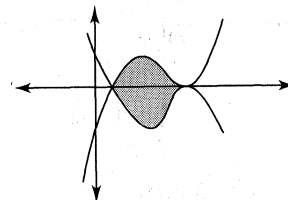


Hence, the required area

$$\begin{aligned}
 &= \int_0^3 (x_2 - x_1) dy \\
 &= \int_0^3 ((y - 2)^2 + 1 - (2y - 4)) dy \\
 &= \int_0^3 (y^2 - 4y + 4 + 1 - 2y + 4) dy \\
 &= \int_0^3 (y^2 - 6y + 9) dy \\
 &= \left(\frac{y^3}{3} - 3y^2 + 9y \right)_0^3 \\
 &= 9 - 27 + 27 \\
 &= 9 \text{ sq.u.}
 \end{aligned}$$

42. Given curve is

$$y^2 = (x - 1)(x - 2)^2$$

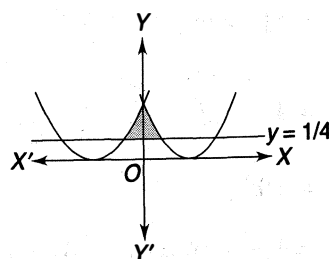


Hence, the required area

$$\begin{aligned}
 &= 2 \int_1^2 (x - 2) \sqrt{x - 1} dx \\
 &= 4 \int_0^1 (t^2 - 1) t^2 dt \\
 &= \left| 4 \int_0^1 (t^4 - t^2) dt \right| \\
 &= \left| 4 \left(\frac{t^5}{5} - \frac{t^3}{3} \right)_0^1 \right| \\
 &= \left| 4 \left(\frac{1}{5} - \frac{1}{3} \right) \right| \\
 &= \frac{8}{15} \text{ sq.u.}
 \end{aligned}$$

43. Given curves are

$$y = (x - 1)^2, y = (x + 1)^2 \text{ and } y = 1/4.$$

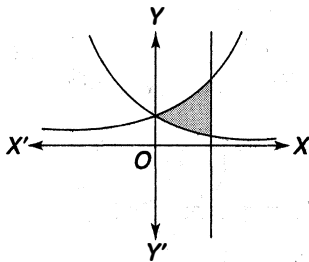


Hence, the required shaded area

$$\begin{aligned}
 &= \int_{-1/2}^0 \left((x+1)^2 - \frac{1}{4} \right) dx + \int_0^{1/2} \left((x-1)^2 - \frac{1}{4} \right) dx \\
 &= \left(\frac{(x+1)^3}{3} - \frac{1}{4}x \right)_{-1/2}^0 + \left(\frac{(x-1)^3}{3} - \frac{1}{4}x \right)_0^{1/2} \\
 &= \frac{1}{3} - \left(\frac{1}{24} + \frac{1}{8} \right) - \frac{1}{24} - \frac{1}{8} + \frac{1}{3} \\
 &= \frac{1}{3} - 2 \left(\frac{1}{24} + \frac{1}{8} \right) + \frac{1}{3} \\
 &= \frac{2}{3} - \frac{2}{6} = \frac{2}{3} - \frac{1}{3} = \frac{1}{3} \text{ sq.u.}
 \end{aligned}$$

44. Given curves are

$y = e^x$, $y = e^{-x}$ and the straight line $x = 1$

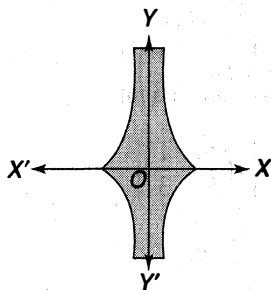


Hence, the required area

$$\begin{aligned}
 &= \int_0^1 (e^x - e^{-x}) dx \\
 &= (e^x + e^{-x})_0^1 \\
 &= \left(e + \frac{1}{e} - 2 \right) \text{ sq.u.}
 \end{aligned}$$

45. Given curves are

$y = \log x$, $y = \log |x|$, $y = |\log x|$ and $y = |\log |x||$

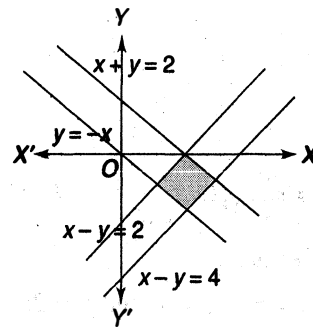


Hence, the required area

$$\begin{aligned}
 &= 4 \int_{-\infty}^0 x dy \\
 &= 4 \int_{-\infty}^0 e^y dy \\
 &= 4 (e^y)_{-\infty}^0 \\
 &= 4 (e^0 - 0) = 4(1 - 0) = 4 \text{ sq.u.}
 \end{aligned}$$

46. Given curve is

$$|x - 2| + |y + 1| = 1$$



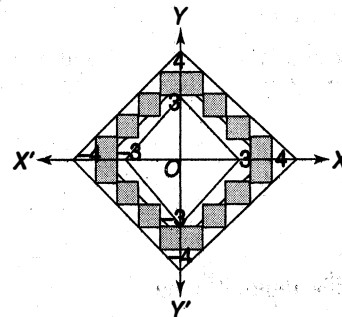
Thus, the length of each side

$$= \sqrt{(2-1)^2 + (0+1)^2} = \sqrt{2}$$

Hence, the required area

$$= (\sqrt{2})^2 \text{ sq.u.}$$

47. Given curve is $[x] + [y] = 3$



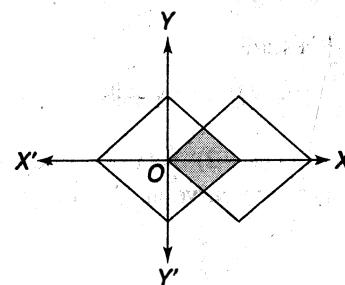
If $[x] = 0, 1, 2, 3$ then $[y] = 3, 2, 1, 0$.

Hence, the required area

$$\begin{aligned}
 &= 4(1 \times 1 + 1 \times 1 + 1 \times 1 + 1 \times 1) \\
 &= 4 \times 4 \\
 &= 16 \text{ sq.u.}
 \end{aligned}$$

48. Given curves are

$$|x| + |y| = 1 \text{ and } |x + 1| + |y| = 1.$$

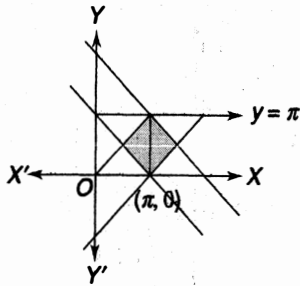


Hence, the required area

$$\begin{aligned}
 &= 2 \times \frac{1}{2} \times 1 \times \frac{1}{2} \\
 &= \frac{1}{2} \text{ sq.u.}
 \end{aligned}$$

49. Given curves are

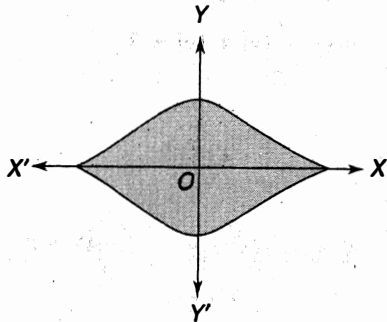
$$y = \cos^{-1}(\cos x) \text{ and } |x - \pi| + \left|y - \frac{\pi}{2}\right| = \frac{\pi}{2}$$



Hence, the required area

$$= 2 \times \frac{1}{2} \times \pi \times \frac{\pi}{2} = \frac{\pi^2}{2} \text{ sq.u.}$$

50. Since the given curve is symmetrical about x -axis as well as y -axis.

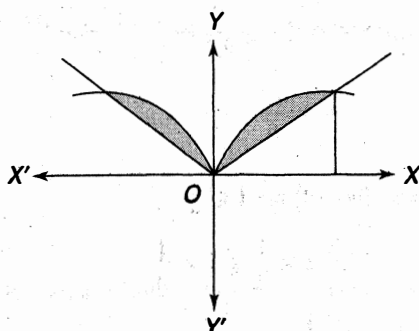


Hence, the required area

$$\begin{aligned} &= 4 \int_0^{\sqrt{5}+1} y dx \\ &= 4 \int_0^{\sqrt{5}+1} (5 - (x-1)^2) dx \\ &= 4 \left(5x - \frac{(x-1)^3}{3} \right)_0^{\sqrt{5}+1} \\ &= \frac{8}{3} (7 + 5\sqrt{5}) \text{ sq.u.} \end{aligned}$$

51. Given curves are

$$y = \sqrt{|x|} \text{ and } y = |x|$$

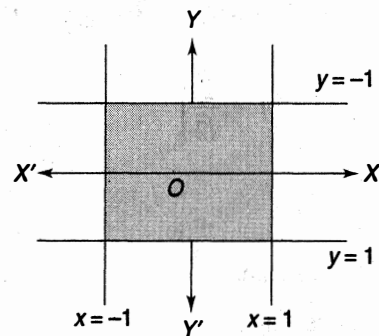


Hence, the required area

$$\begin{aligned} &= 2 \int_0^1 (\sqrt{x} - x) dx \\ &= 2 \left(\frac{2}{3} x^{3/2} - \frac{x^2}{2} \right)_0^1 \\ &= 2 \left(\frac{2}{3} - \frac{1}{2} \right) \\ &= 2 \left(\frac{4-3}{6} \right) = 2 \times \frac{1}{6} = \frac{1}{3} \text{ sq.u.} \end{aligned}$$

52. Given curve is $|x + y| + |x - y| \leq 2$

$$= \begin{cases} x \leq 1, x \geq -1 \\ y \leq 1, y \geq -1 \end{cases}$$



Hence, the required area

$$= (2)^2 = 4 \text{ sq.u.}$$

53. Given curves are

$$y = x, y = x + \sin x, 0 \leq x \leq \pi$$

Hence, the required area

$$\begin{aligned} &= \int_0^\pi (x \sin x - x) dx \\ &= \int_0^\pi (\sin x) dx \\ &= -(\cos x)_0^\pi = -(-1 - 1) = 2 \text{ sq.u.} \end{aligned}$$

54. Hence, the required area

$$= \int_{\frac{\pi}{4}}^\beta f(x) dx = \left(\beta \sin \beta + \frac{\pi}{4} \cos \beta + \sqrt{2} \beta \right)$$

Applying, Newton and Leibnitz rule, we get

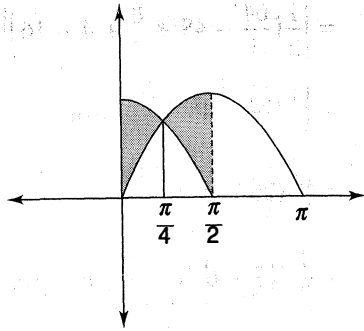
$$f(\beta) = \beta \cos \beta + \sin \beta - \frac{\pi}{4} \sin \beta + \sqrt{2}$$

$$f\left(\frac{\pi}{2}\right) = 0 + 1 - \frac{\pi}{4} + \sqrt{2}$$

$$f\left(\frac{\pi}{2}\right) = \left(1 - \frac{\pi}{4}\right) + \sqrt{2}$$

55. Given curves are $y = \sin x$ and $y = \cos x$

$$x = 0 \text{ and } x = \frac{\pi}{2}$$

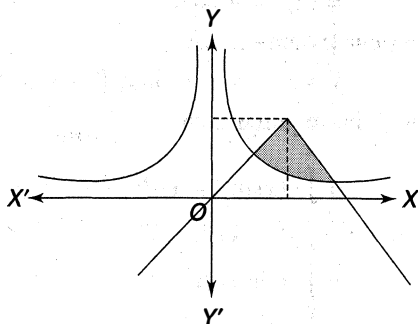


Hence, the required area

$$\begin{aligned} & \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx \\ &= (\sin x + \cos x) \Big|_0^{\pi/4} - (\cos x + \sin x) \Big|_{\pi/4}^{\pi/2} \\ &= \left(\frac{2}{\sqrt{2}} - 1 \right) - \left(1 - \frac{2}{\sqrt{2}} \right) \\ &= 2 \left(\frac{2}{\sqrt{2}} - 1 \right) \\ &= 2(\sqrt{2} - 1) \text{ sq.u.} \end{aligned}$$

56. Given curves are

$$y = 2 - |2 - x| \text{ and } y = \frac{3}{|x|}$$



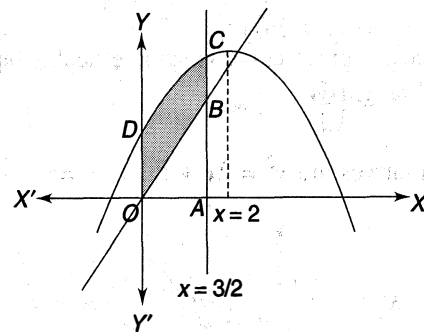
Hence, the required area

$$\begin{aligned} &= \int_{\sqrt{3}}^2 \left(x - \frac{3}{x} \right) dx + \int_2^{2+\sqrt{7}} \left(4 - x - \frac{3}{x} \right) dx \\ &= \left(\frac{x^2}{2} - 3 \log x \right) \Big|_{\sqrt{3}}^2 + \left(4x - \frac{x^2}{2} - 3 \log x \right) \Big|_2^{2+\sqrt{7}} \\ &= \left(2 - 3 \log 2 - \frac{3}{2} + \frac{3}{2} \log 2 \right) \\ &\quad + \left(4(2 + \sqrt{7}) - \frac{(2 + \sqrt{7})^2}{2} - 3 \log(2 + \sqrt{7}) \right) \\ &\quad - 8 + 2 + 3 \log 2 \end{aligned}$$

$$\begin{aligned} &= \left(\frac{1}{2} + \frac{3}{2} \log 2 \right) + 3 \log \left(\frac{-2}{2 + \sqrt{7}} \right) + \left(2\sqrt{7} - \frac{11}{2} \right) \\ &= \left(2\sqrt{7} + \frac{3}{2} \log 2 - 5 \right) + 3 \log \left(\frac{-2}{2 + \sqrt{7}} \right) \\ &= (2\sqrt{7} - 5) + 3 \log \left(\frac{2}{2 + \sqrt{7}} \right) + 3 \log(\sqrt{2}) \\ &= (2\sqrt{7} - 5) + 3 \log \left(\frac{2\sqrt{2}}{2 + \sqrt{7}} \right) \end{aligned}$$

57. Given curve is $y = 4x - x^2$

$$\Rightarrow y = 5 - (x - 2)^2$$



Now, the area $OABO$

$$\begin{aligned} &= \int_0^{3/2} (mx) dx \\ &= \left(\frac{mx^2}{2} \right) \Big|_0^{3/2} = \left(\frac{9m}{8} \right) \end{aligned}$$

Hence, the area $OABO$

$$\begin{aligned} &= \int_0^{3/2} (1 + 4x - x^2) dx \\ &= \left(x + 2x^2 - \frac{x^3}{3} \right) \Big|_0^{3/2} \\ &= \left(\frac{3}{2} + 2 \cdot \frac{9}{4} - \frac{27}{24} \right) \\ &= \left(6 - \frac{9}{8} \right) = \frac{39}{8} \text{ sq.u} \end{aligned}$$

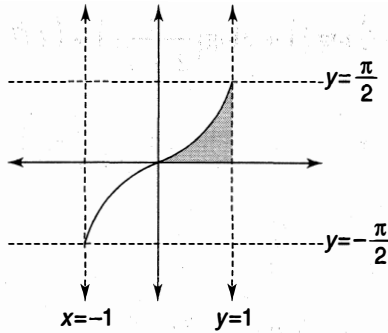
According to the question, $\frac{9m}{8} = \frac{1}{2} \cdot \frac{39}{8}$

$$\Rightarrow m = \frac{13}{6}$$

Hence, the value of m is $\frac{13}{6}$.

58. Do yourself.

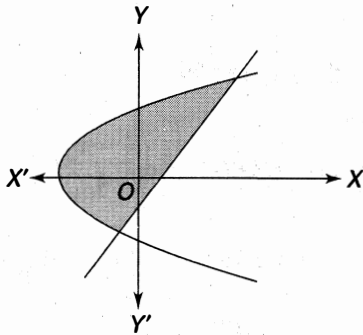
59. Given curve is $y = \sin^{-1} x$ and y -axis between $y = 0$ and $y = \pi/2$.



Hence, the required area

$$\begin{aligned} &= \int_0^{\frac{\pi}{2}} (1 - \sin x) dx \\ &= (y + \cos y)_0^{\frac{\pi}{2}} \\ &= \left(\frac{\pi}{2} - 1\right) \text{ sq.u.} \end{aligned}$$

60. Given curves are $y^2 = 2x + 6$ and $y = x - 1$



On solving, we get, $(x - 1)^2 = 2x + 6$

$$y^2 = 2(y + 1) + 6$$

$$y^2 - 2y - 8 = 0$$

$$(y - 4)(y + 2) = 0$$

$$y = -2, 4$$

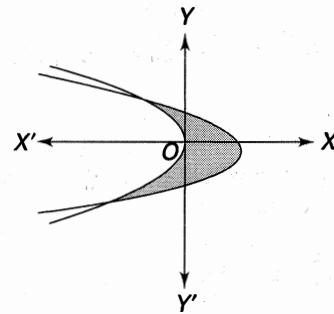
Hence, the required area

$$\begin{aligned} &= \int_{-2}^4 (x_2 - x_1) dy \\ &= \int_{-2}^4 \left(\frac{y^2 - 6}{2} - (y + 1)\right) dy \\ &= \frac{1}{2} \int_{-2}^4 (y^2 - 6 - 2y - 2) dy \\ &= \frac{1}{2} \int_{-2}^4 (y^2 - 2y - 8) dy \\ &= \frac{1}{2} \left(\frac{y^3}{3} - y^2 - 8y\right)_{-2}^4 \end{aligned}$$

$$\begin{aligned} &= \left| \frac{1}{2} \left(\frac{64}{3} - 48 + \frac{8}{3} + 4 - 16\right) \right| \\ &= \left| \frac{1}{2} (24 - 48 + 4 - 16) \right| \\ &= \left| \frac{1}{2} (28 - 64) \right| \\ &= \frac{1}{2} |(28 - 64)| = \frac{36}{2} = 18 \text{ sq.u.} \end{aligned}$$

61. Given curves are

$$x = -2y^2 \text{ and } x = 1 - 3y^2.$$



On solving, we get,

$$-2y^2 = 1 - 3y^2$$

$$y^2 = 1$$

$$y = \pm 1$$

Hence, the required area

$$\begin{aligned} &= \int_{-1}^1 (x_2 - x_1) dy \\ &= \int_{-1}^1 ((-2y^2) - (1 - 3y^2)) dy \\ &= \int_{-1}^1 (y^2 - 1) dy \\ &= 2 \int_0^1 (y^2 - 1) dy \\ &= 2 \left(\frac{y^3}{3} - y\right)_0^1 \\ &= 2 \left|\left(\frac{1}{3} - 1\right)\right| = \frac{4}{3} \text{ sq.u.} \end{aligned}$$

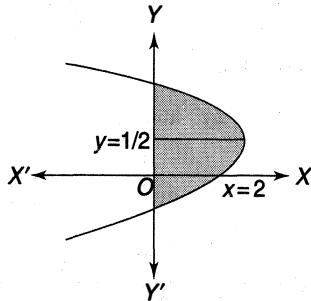
62. Given curves are $y = \tan^{-1}x$ and $y = \cot^{-1}x$ and the y-axis.

Hence, the required area

$$\begin{aligned} &= \int_0^{\frac{\pi}{4}} \tan y dy + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot y dy \\ &= (\log(\sec y))_0^{\frac{\pi}{4}} + (\log(\sin y))_{\frac{\pi}{4}}^{\frac{\pi}{2}} \end{aligned}$$

$$\begin{aligned}
 &= \log(\sqrt{2}) + \left(0 - \log\left(\frac{1}{\sqrt{2}}\right)\right) \\
 &= 2\log(\sqrt{2}) \\
 &= \log(2) \text{ sq.u.}
 \end{aligned}$$

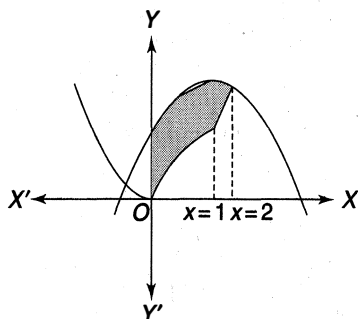
63. Given curve is $x = y^2 + y + 2$ and the y-axis.



Hence, the required area

$$\begin{aligned}
 &= \int_{-1}^2 x dy \\
 &= \int_{-1}^2 (-y^2 + y + 2) dy \\
 &= \left(-\frac{y^3}{3} + \frac{y^2}{2} + 2y\right)_{-1}^2 \\
 &= \left(-\frac{8}{3} + 2 + 4 + \frac{1}{3} - \frac{1}{2} - 2\right) \\
 &= \left(4 - \frac{7}{3} - \frac{1}{2}\right) = \frac{7}{6} \text{ sq.u.}
 \end{aligned}$$

64. Given curves are $y = \sqrt{x}$, $y = x^2$
 $y = -(x^2 - 2x - 4) = 5 - (x - 1)^2$



Hence, the required area

$$\begin{aligned}
 &= \int_0^2 (-x^2 + 2x + 4) dx - \int_0^1 \sqrt{x} dx - \int_1^2 x^2 dx \\
 &= \left(-\frac{x^3}{3} + x^2 + 4x\right)_0^2 - \left(\frac{2}{3}x^{3/2}\right)_0^1 - \left(\frac{x^3}{3}\right)_1^2 \\
 &= \left(-\frac{18}{3} + 12\right) - \left(\frac{2}{3}\right) - \left(\frac{8}{3} - \frac{1}{3}\right)
 \end{aligned}$$

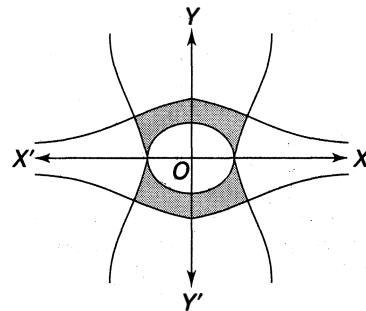
$$\begin{aligned}
 &= \frac{28}{3} - \frac{2}{3} - \frac{7}{3} \\
 &= \frac{28 - 9}{3} \\
 &= \frac{19}{3} \text{ sq.u.}
 \end{aligned}$$

65. Find the area of the region bounded by the curves

$$4|y| = |4 - x^2| \text{ and } |y|(x^2 + 4) = 12.$$

65. Given curves are

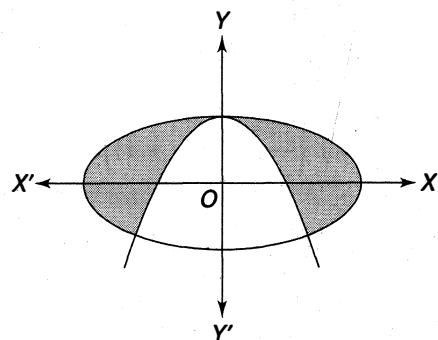
$$4|y| = |4 - x^2| \text{ and } |y|(x^2 + 4) = 12.$$



Hence, the required area

$$\begin{aligned}
 &= 4 \left[\int_0^{2\sqrt{2}} \left(\frac{12}{x^2 + 4}\right) dx - \left(\int_0^2 \left(\frac{4 - x^2}{4}\right) dx \right) \right. \\
 &\quad \left. + \left(\int_2^{2\sqrt{2}} \left(\frac{x^2 - 4}{4}\right) dx \right) \right] \\
 &= 4 \left[\left(6 \tan^{-1}\left(\frac{x}{2}\right)\right)_0^{2\sqrt{2}} - \left(x - \frac{x^3}{12}\right)_0^2 - \left(\frac{x^3}{12} - x\right)_2^{2\sqrt{2}} \right] \\
 &= 4 \left[6 \tan^{-1}(\sqrt{2}) - \frac{4}{3} + \frac{4 - 2\sqrt{2}}{3} \right] \\
 &= \frac{4}{3} (18 \tan^{-1}(\sqrt{2}) - 2\sqrt{2}) \text{ sq.u.}
 \end{aligned}$$

66. Given curves are $x^2 + 2y^2 = 2$ and $y = 1 - x^2$

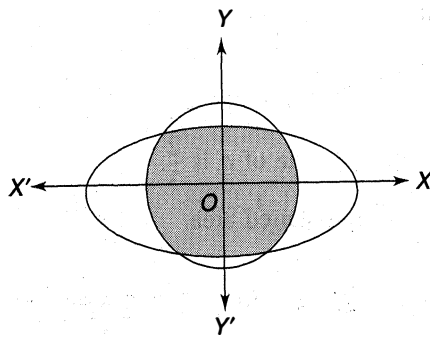


Hence, the required area

$$= 2 \int_{-1/2}^1 (\sqrt{2 - 2y^2} - \sqrt{1 - y}) dy$$

$$\begin{aligned}
 &= 2\sqrt{2} \int_{-1/2}^1 \sqrt{1-y^2} dy - 2 \int_{-1/2}^1 \sqrt{1-y} dy \\
 &= 2\sqrt{2} \left(\frac{y}{2} \sqrt{1-y^2} + \frac{1}{2} \sin^{-1}(y) \right)_{-1/2}^1 \\
 &\quad - 2 \left(\frac{2}{-3} (1-y)^{3/2} \right)_{-1/2}^1 \\
 &= 2\sqrt{2} \left(\frac{\pi}{4} + \frac{1}{4} \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \sin^{-1}\left(\frac{1}{2}\right) \right) - \frac{4}{3} \left(\frac{3}{2} \right)^{3/2} \\
 &= 2\sqrt{2} \left(\frac{\pi}{4} + \frac{\sqrt{3}}{8} + \frac{\pi}{12} \right) - \frac{4}{3} \left(\frac{3\sqrt{3}}{2\sqrt{2}} \right) \\
 &= 2\sqrt{2} \left(\frac{\pi}{3} + \frac{\sqrt{3}}{8} \right) - 2 \left(\frac{\sqrt{3}}{\sqrt{2}} \right) \\
 &= \left(2\sqrt{2} \left(\frac{\pi}{3} + \frac{\sqrt{3}}{8} \right) - \sqrt{6} \right) \text{ sq.u.}
 \end{aligned}$$

67. Given curves are $x^2 + y^2 = 4$ and $x^2 + 4y^2 = 9$



Solving the above equations, we get,

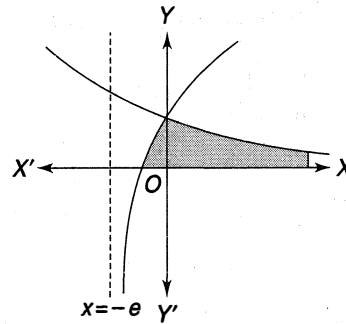
$$\begin{aligned}
 &x^2 + 4(4 - x^2) = 9 \\
 \Rightarrow &3x^2 = 7 \\
 \Rightarrow &x = \sqrt{\frac{7}{3}}
 \end{aligned}$$

Hence, the required area

$$\begin{aligned}
 &= 4 \left(\int_0^{\sqrt{7/3}} \frac{1}{2} \sqrt{9-x^2} dx + \int_{\sqrt{7/3}}^2 \sqrt{4-x^2} dx \right) \\
 &= 4 \left[\frac{1}{2} \left(\frac{x}{2} \sqrt{9-x^2} + \frac{9}{2} \sin^{-1}\left(\frac{x}{3}\right) \right)_{\sqrt{7/3}}^2 \right. \\
 &\quad \left. + \frac{x}{2} \sqrt{4-x^2} + 2 \sin^{-1}\left(\frac{x}{2}\right) \right]_{\sqrt{7/3}}^2 \\
 &= \left\{ 4\pi + 9 \sin^{-1}\left(\frac{1}{3} \sqrt{\frac{7}{3}}\right) - 8 \sin^{-1}\left(\frac{1}{2} \sqrt{\frac{7}{3}}\right) \right\}
 \end{aligned}$$

68. Given curves are

$$\begin{aligned}
 &y = \ln(x+e) \text{ and } x = \ln\left(\frac{1}{y}\right) \\
 \Rightarrow &y = \ln(x+e), y = e^{-x}, y = 0
 \end{aligned}$$

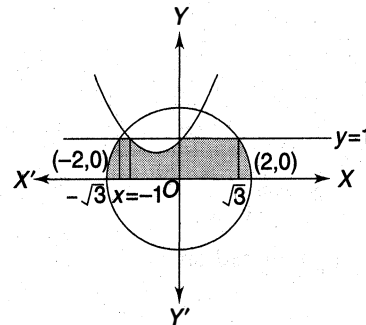


Hence, the required area

$$\begin{aligned}
 &= \int_{1-e}^0 \log(x+e) dx + \int_0^\infty e^{-x} dx \\
 &= ((x+e) \log(x+e) - (x+e))_{1-e}^0 - (e^{-x})_0^\infty \\
 &= (e \log e - e) - (1 \log 1 - 1) - (e^{-x})_0^\infty \\
 &= (1 - (0 - 0)) \\
 &= 2 \text{ sq.u.}
 \end{aligned}$$

69. Given curves are $x^2 + y^2 = 4$, $y = x^2 + x + 1$ and

$$y = \left[\sin^2\left(\frac{x}{4}\right) + \cos\left(\frac{x}{4}\right) \right] \text{ and } y = 0.$$

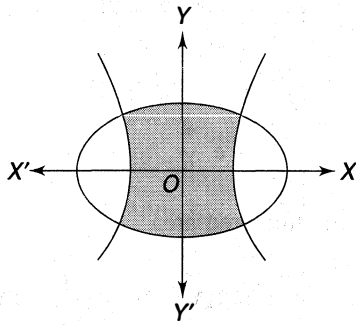


Hence, the required area

$$\begin{aligned}
 &= (\sqrt{3} \times 1) + (\sqrt{3} - 1) \\
 &\quad + \int_{-1}^0 (x^2 + x + 1) dx + 2 \int_{\sqrt{3}}^2 \sqrt{4-x^2} dx \\
 &= (2\sqrt{3} - 1) + \left(\frac{x^3}{3} + \frac{x^2}{2} + x \right)_{-1}^0 \\
 &\quad + 2 \left(\frac{x}{2} \sqrt{4-x^2} + 2 \sin^{-1}\left(\frac{x}{2}\right) \right)_{\sqrt{3}}^2 \\
 &= (2\sqrt{3} - 1) + \left(0 - \left(-\frac{1}{3} + \frac{1}{2} - 1 \right) \right) \\
 &\quad + 2 \left\{ (0 + \pi) - \left(\frac{\sqrt{3}}{2} + \frac{2\pi}{3} \right) \right\} \\
 &= \left\{ (2\sqrt{3} - 1) + \frac{5}{6} + \frac{2\pi}{3} - \sqrt{3} \right\} \\
 &= \left(\frac{2\pi}{3} + \sqrt{3} - \frac{1}{6} \right) \text{ sq.u.}
 \end{aligned}$$

70. Given curves are

$$\frac{x^2}{4} + y^2 = 1 \text{ and } \frac{x^2}{2} - y^2 = 1$$



Solving the above equations, we get

$$\frac{x^2}{4} + \frac{x^2}{2} = 2$$

$$\Rightarrow \frac{3x^2}{4} = 2$$

$$\Rightarrow x^2 = \frac{8}{3}$$

$$\text{Now, } y^2 = 1 - \frac{x^2}{4}$$

$$\Rightarrow y^2 = 1 - \frac{8}{4 \times 3} = 1 - \frac{2}{3} = \frac{1}{3}$$

$$\Rightarrow y = \frac{1}{\sqrt{3}}$$

Hence, the required area

$$\begin{aligned} &= 4 \left(\int_0^{1/\sqrt{3}} (\text{Hyperbola}) dy + \int_{1/\sqrt{3}}^1 (\text{Ellipse}) dy \right) \\ &= 4 \left(\int_0^{1/\sqrt{3}} \sqrt{2}(\sqrt{1+y^2}) dy + \int_{1/\sqrt{3}}^1 2\sqrt{1+y^2} dy \right) \\ &= 4 \left[\sqrt{2} \left(\frac{y}{2} \sqrt{1+y^2} + \frac{1}{2} \log|y + \sqrt{y^2+1}| \right) \right]_0^{1/\sqrt{3}} \\ &\quad + 2 \left(\frac{y}{2} \sqrt{1+y^2} + \frac{1}{2} \log|y + \sqrt{y^2+1}| \right) \Big|_{1/\sqrt{3}}^1 \\ &= 4 \left[\sqrt{2} \left(\frac{1}{2\sqrt{3}} \times \frac{2}{\sqrt{3}} + \frac{1}{2} \log \left| \frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} \right| \right) \right. \\ &\quad \left. + 2 \left(\frac{1}{\sqrt{2}} + \frac{1}{2} \log(\sqrt{2}+1) - \frac{1}{2\sqrt{3}} \times \frac{2}{\sqrt{3}} - \frac{1}{4} \log 3 \right) \right] \\ &= 4 \left[\sqrt{2} \left(\frac{1}{3} + \frac{1}{4} \log(3) \right) \right] \end{aligned}$$

$$+ 2 \left(\frac{1}{\sqrt{2}} + \frac{1}{2} \log(\sqrt{2}+1) - \frac{1}{3} - \frac{1}{4} \log 3 \right) \Big] \text{ sq.u.}$$

71. Let the area be A.

$$\begin{aligned} \text{Then } A &= \int_0^2 \left(\frac{x^3}{3} - x^2 + a \right) dx \\ &= \left(\frac{x^4}{12} - \frac{x^3}{3} + ax \right)_0^2 \\ &= \left(\frac{4}{3} - \frac{8}{3} + 2a \right) \\ &= \left(2a - \frac{4}{3} \right) \end{aligned}$$

$$\frac{dA}{da} = 2 > 0$$

So, the area is minimum.

It will be minimum when $A = 0$.

$$\text{Thus, } 2a - \frac{4}{3} = 0$$

$$\Rightarrow a = \frac{2}{3}$$

Hence, the value of a is $\frac{2}{3}$.

72. Hence, the required area

$$\begin{aligned} &= A = \int_0^1 (a^2x^2 + ax + 1) dx \\ &= \left(\frac{a^2x^3}{3} + \frac{ax^2}{2} + x \right)_0^1 \\ &= \left(\frac{a^2}{3} + \frac{a}{2} + 1 \right) \\ \Rightarrow \frac{dA}{da} &= \frac{2a}{3} + \frac{1}{2} \\ \Rightarrow \frac{d^2A}{da^2} &= \frac{2}{3} > 0 \end{aligned}$$

So, the area is minimum.

$$\text{For maximum or minimum, } \frac{2a}{3} + \frac{1}{2} = 0$$

$$\Rightarrow \frac{2a}{3} = -\frac{1}{2}$$

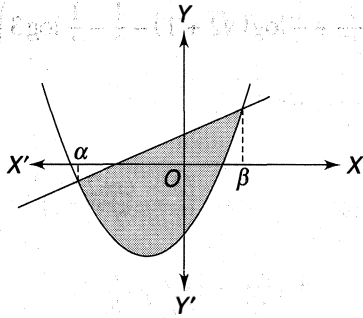
$$\Rightarrow a = -\frac{3}{4}$$

Hence, the value of a is $-3/4$

73. Do yourself.

74. Given curves are

$$y = x^2 + 2x - 3 \text{ and } y = mx + 1$$



Solving, we get

$$x^2 + 2x - 3 = mx + 1$$

$$x^2 + (2 - m)x - 4 = 0$$

Let α, β be the roots such that

$$\alpha + \beta = m - 2, \alpha\beta = -4$$

Hence, the required area

$$\begin{aligned} A &= \int_{\alpha}^{\beta} (\text{Line} - \text{Parabola}) dx \\ &= \int_{\alpha}^{\beta} ((mx + 1) - (x^2 + 2x - 3)) dx \\ &= \left(\frac{mx^2}{2} + x - \frac{x^3}{3} - x^2 + 3x \right)_{\alpha}^{\beta} \\ &= \left(\left(\frac{m\beta^2}{2} + 4\beta - \frac{\beta^3}{3} - \beta^2 \right) - \left(\frac{m\alpha^2}{2} + 4\alpha - \frac{\alpha^3}{3} - \alpha^2 \right) \right) \\ &= \left(\left(\frac{m}{2}(\beta^2 - \alpha^2) + 4(\beta - \alpha) - \frac{(\beta^3 - \alpha^3)}{3} - (\beta^2 - \alpha^2) \right) \right) \\ &= \left(\left(\left(\frac{m}{2} - 1 \right)(\beta^2 - \alpha^2) + 4(\beta - \alpha) - \frac{(\beta^3 - \alpha^3)}{3} \right) \right) \\ &= (\beta - \alpha) \left(\left(\frac{m}{2} - 1 \right)(\beta + \alpha) + 4 - \frac{(\beta^2 + \alpha\beta + \alpha^2)}{3} \right) \\ &= (\beta - \alpha) \left(\frac{(m - 2)^2}{2} + 4 - \frac{(m - 2)^2 + 4}{3} \right) \\ &= (\beta - \alpha) \left(\frac{(m - 2)^2}{6} + \frac{16}{3} \right) \\ &= \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} \left(\frac{(m - 2)^2}{6} + \frac{16}{3} \right) \\ &= \sqrt{(m - 2)^2 + 16} \left(\frac{(m - 2)^2}{6} + \frac{16}{3} \right) \end{aligned}$$

The area is minimum, if $m = 2$ and the least area is

$$= \frac{64}{3}$$

75. Hence, the required area

$$\begin{aligned} &= A = \int_a^{2a} \left(\frac{x}{6} + \frac{1}{x^2} \right) dx \\ &= \left(\frac{x^2}{12} - \frac{1}{x} \right)_{a}^{2a} \\ &= \left(\frac{a^2}{3} - \frac{1}{2a} - \frac{a^2}{12} + \frac{1}{a} \right) \\ &= \left(\frac{a^2}{4} + \frac{1}{2a} \right) \end{aligned}$$

$$\frac{dA}{da} = \frac{a}{2} - \frac{1}{2a^2}$$

$$\Rightarrow \frac{d^2A}{da^2} = \frac{1}{2} + \frac{1}{a^3}$$

Now for maximum or minimum,

$$\frac{dA}{da} = 0$$

$$\Rightarrow \frac{a}{2} - \frac{1}{2a^2} = 0$$

$$\Rightarrow \frac{a}{2} = \frac{1}{2a^2}$$

$$\Rightarrow a^3 = 1$$

$$\Rightarrow a = 1$$

Hence, the value of a is 1, when the area is least.